Acantholytic Blistering of Oral and Laryngeal Mucosa (ABOLM) — Comprehensive Disease Characterization Report

Target disease: Acantholytic Blistering of Oral and Laryngeal Mucosa (ABOLM) **MONDO ID:** MONDO:0030986 · **Category:** Mendelian (monogenic, autosomal recessive) **Causal gene:** *DSG3* (desmoglein 3), 18q12.1, HGNC:3050, OMIM gene *605595*

Summary

Acantholytic Blistering of Oral and Laryngeal Mucosa (ABOLM) is an ultra-rare, autosomal-recessive, non-autoimmune mucosal blistering disorder caused by biallelic loss-of-function of *DSG3* (desmoglein 3). It is, in essence, the *human genetic null* equivalent of the autoimmune disease pemphigus vulgaris (PV): the target antigen that is attacked by autoantibodies in PV is simply absent from birth in ABOLM. The disorder has been reported in only two families worldwide — a consanguineous family with a homozygous *DSG3* nonsense mutation (PMID: 30528827, 2019) and dizygotic twins with a homozygous 49.6-kb deletion of chromosome 18q12.1 removing the terminal exon of *DSG3* (PMID: 37850634, 2024). As of this report, it is one of the rarest monogenic desmosomal diseases described.

The central mechanism is loss of desmoglein-3–mediated cell–cell adhesion in stratified squamous epithelia. DSG3 is a calcium-dependent desmosomal cadherin whose N-terminal (EC1) adhesive interface mediates *trans*-adhesion between adjacent keratinocytes. When DSG3 is absent, desmosomes in the deep (suprabasal/basal) layers of mucosal epithelium fail, producing **suprabasal acantholysis** — the same histology ("tombstone" basal cells) seen in pemphigus vulgaris. Critically, the disease is **mucosa-dominant and largely skin-sparing**, a pattern explained by the **desmoglein compensation theory**: in skin, DSG1 is broadly expressed and compensates for DSG3 loss, whereas in oral and laryngeal mucosa DSG1 expression is weak and cannot compensate, so DSG3 loss produces full mucosal blistering. The

Dsg3-knockout mouse (and its coallelic spontaneous mutant "balding") faithfully recapitulates the oral acantholytic phenotype, runting from impaired feeding, and cyclical telogen hair loss, providing strong cross-species validation.

For a knowledge base, the actionable clinical message is diagnostic and therapeutic distinction from pemphigus. ABOLM shows PV-like histology but **negative** direct/indirect immunofluorescence and negative DSG1/DSG3 autoantibody ELISA; molecular confirmation is biallelic *DSG3* loss-of-function. Because the disorder is non-autoimmune, the immunosuppressive therapies that anchor pemphigus management (corticosteroids, rituximab) are **not indicated**; care is **supportive**, focused on airway protection, nutrition, pain control, and avoidance of mucosal trauma. The 2024 twins presented predominantly with laryngeal/respiratory mucosal friability, underscoring airway involvement as a defining and clinically important feature.

Key Findings

Finding 1 — ABOLM is caused by biallelic loss-of-function of *DSG3*

ABOLM is an autosomal-recessive mucosal blistering disorder caused by complete loss of desmoglein 3 function from both *DSG3* alleles. Two independent human reports establish causation across two distinct mutational mechanisms. The first family carried a **homozygous nonsense mutation** in *DSG3* (PMID: 30528827, 2019). The second report described a **homozygous 49.6-kb deletion of chromosome 18q12.1** removing the last exon of *DSG3* in dizygotic twins (PMID: 37850634, 2024). The authors of the 2024 report state: *"We report a novel homozygous 49.6 kb deletion of chromosome 18q12.1 involving the last exon of DSG3 in dizygotic twins with phenotype consistent with acantholytic blistering of the oral and laryngeal mucosa (ABOLM),"* and note this is *"only the second report in the literature of this unusual autosomal recessive blistering disorder."* The phenotype is acantholytic blistering and erosions of the oral and laryngeal/respiratory mucosa **without autoantibodies** — *"The diagnosis explains the mucosal phenotype of a pemphigus-like disorder without evidence of autoimmune dysfunction."* Two convergent loss-of-function mechanisms (nonsense; terminal-exon deletion) producing the same phenotype strongly support *DSG3* null as the definitive cause. **Evidence type: human clinical/genetic.**

Finding 2 — The *Dsg3*-null mouse (and coallelic "balding" mutant) recapitulates the disease

The *DSG3*-knockout mouse provides a faithful animal model and demonstrates the direct adhesive function of DSG3. The authors engineered mice with targeted disruption of *DSG3* specifically *"to test whether Dsg3 mediates adhesion"* (PMID: 9166409). *DSG3*−/− mice are normal at birth but become runted around day 18 as oral lesions impair feeding. Oropharyngeal biopsies reproduce the human histology: *"oropharyngeal biopsies showed erosions with histology typical of PV, including suprabasilar acantholysis and 'tombstoning' of basal cells. EM showed separation of desmosomes."* The phenotype is **identical to the spontaneous mouse mutant "balding" (bal)**: *"The runting and hair loss phenotype of DSG3 −/− mice is identical to that of a previously reported mouse mutant, balding (bal),"* establishing *bal* as a naturally occurring *DSG3*-null model. This cross-species concordance — same gene, same suprabasal acantholytic histology, same desmosomal ultrastructural separation — is a cornerstone of the causal argument. **Evidence type: model organism (mouse).**

Finding 3 — The desmoglein compensation theory explains mucosa-dominant, skin-sparing disease

Why does loss of a protein expressed throughout stratified epithelia produce disease predominantly in mucosa and hair while largely sparing skin? The **desmoglein compensation theory** (Stanley & Amagai) provides the mechanism. DSG1 and DSG3 have complementary tissue distributions. In oral mucosa, DSG3 is expressed throughout the epithelium and DSG1 is weak and limited, so DSG3 loss cannot be compensated and acantholysis results. In skin, DSG1 is expressed across all layers and compensates for DSG3 loss in the superficial layers, sparing most skin (acantholysis occurs only with added mechanical trauma). The theory was established through anti-desmoglein autoantibody profiling in pemphigus: *"The desmoglein compensation theory posits that this rare phenotype can be produced by pathogenically weak anti-Dsg3 IgG in the presence of potent anti-Dsg1 IgG autoantibodies"* (PMID: 15858475). It explains lesion localization directly: *"the Dsg compensation theory proposed by Stanley and Amagai can reasonably explain the different depths of skin lesions and the different occurrences of skin and oral mucosal lesions between PV and PF"* (PMID: 12677426). Applied to ABOLM, the complete genetic absence of DSG3 is phenotypically equivalent to the "mucosal-dominant" autoantibody profile of PV, but constitutively and from birth. **Evidence type: human clinical + mechanistic inference.**

Finding 4 — DSG3 adhesion operates through a calcium-dependent N-terminal (EC1) interface

The molecular basis of DSG3-mediated adhesion is a calcium-dependent, conformational, N-terminal *trans*-adhesive interface, and its disruption alone is sufficient to cause acantholysis. The pathogenic monoclonal antibody **AK23** reproduces the full *Dsg3*−/−/PV phenotype by binding this interface: *"AK23 recognized a calcium-dependent conformational epitope on Dsg3, which consisted of the V3, K7, P8, and D59 Dsg3-specific residues that formed the adhesive interface between juxtaposed Dsg, as predicted by the crystal structure"* (PMID: 12574390). Antibodies binding middle or C-terminal extracellular-cadherin (EC) regions are non-pathogenic, localizing the critical adhesive function to the EC1 amino terminus. The same report concludes these data *"suggest the direct inhibition of adhesive interaction of Dsg as an initial molecular event of blister formation in pemphigus."* DSG3 is a type-1 transmembrane desmosomal cadherin with four extracellular cadherin domains (EC1–EC4) plus an EC-anchor domain, a single transmembrane segment, and a cytoplasmic tail that binds plakoglobin. In ABOLM, genetic ablation removes the entire protein — including this interface — so the proximal adhesive lesion is identical to (and more complete than) antibody-mediated interface blockade. **Evidence type: in vitro / structural / model organism.**

Finding 5 — DSG3 anchors telogen-phase hair; its loss causes cyclical acantholytic hair loss

Beyond mucosa, DSG3 has a specialized role anchoring resting-phase (telogen) hair in the follicle, explaining the hair phenotype of the null state. In *DSG3*−/− mice, hair grows normally through the first anagen (growth) cycle, but *"when the hair follicles entered the resting phase of the hair growth cycle (telogen), mice with a targeted disruption of the desmoglein 3 gene (DSG3−/−) lost hair in a wave-like pattern from the head to the tail"* (PMID: 9701552). Telogen (not anagen) hairs pull out in clumps. Histology shows cystic telogen follicles lacking shafts, and the mechanism is again acantholytic: the study *"showed loss of cell adhesion (acantholysis) between the cells surrounding the telogen club and the basal layer of the outer root sheath epithelium,"* with "half-desmosomes" on EM. This generalizes the DSG3 adhesion defect from interfollicular mucosa to the follicular outer root sheath. (Human ABOLM reports emphasize the mucosal/airway phenotype; hair involvement in humans is less documented and may be subtler, but the mouse clarifies the full biological role of DSG3.) **Evidence type: model organism (mouse).**

Finding 6 — Diagnosis: suprabasal acantholysis + NEGATIVE immunofluorescence/serology, confirmed genetically; management is supportive

The diagnostic signature of ABOLM is PV-like histology combined with the *absence* of autoimmune markers, confirmed by *DSG3* genetic testing. Biopsy shows suprabasilar acantholysis and "tombstone" basal cells, but — unlike pemphigus — there are **no circulating or tissue-bound anti-desmoglein autoantibodies**: direct immunofluorescence (DIF), indirect immunofluorescence (IIF), and DSG1/DSG3 ELISA are all negative. Molecular confirmation is biallelic *DSG3* loss-of-function (homozygous nonsense, PMID: 30528827; homozygous 18q12.1 deletion detectable by chromosomal microarray/sequencing, PMID: 37850634). This distinction is not academic: *"The exclusion of an autoimmune basis has management implications"* (PMID: 37850634) — meaning immunosuppressants and rituximab used for pemphigus are not indicated, and care is supportive. Airway management is central because *"The twin siblings presented predominantly with friability of the laryngeal and respiratory mucosa."* **Evidence type: human clinical.**

The 15-Section Disease Profile

1. Disease Information

ABOLM is a Mendelian, autosomal-recessive, non-autoimmune mucosal blistering disorder in which biallelic *DSG3* loss-of-function abolishes desmoglein-3–mediated keratinocyte adhesion, producing suprabasal acantholytic blisters and erosions of the oral and laryngeal/respiratory mucosa. It is essentially a genetic phenocopy of pemphigus vulgaris lacking the autoimmune component.

Identifier type	Value
MONDO	MONDO:0030986
Gene / locus	*DSG3*, 18q12.1
HGNC	HGNC:3050
OMIM (gene)	*605595* (DSG3)
ICD	No specific code; maps under non-autoimmune acantholytic/blistering mucosal disorders
MeSH	No dedicated descriptor; related: Pemphigus; Desmoglein 3; Acantholysis

Synonyms / alternative names: Acantholytic blistering of oral and laryngeal mucosa; DSG3-related mucosal blistering; hereditary (non-autoimmune) pemphigus-like mucosal blistering; human desmoglein-3 deficiency. **Information source:** aggregated disease-level resources and individual case reports (two families). This is a case-report–level disease, not an EHR/aggregate-cohort entity.

2. Etiology

Primary cause: genetic — biallelic loss-of-function variants in *DSG3*. No environmental, infectious, or immune trigger is required; the disease is constitutive. **Genetic risk factor** = inheriting two null *DSG3* alleles (autosomal recessive); **consanguinity** is a contributing factor (the homozygous nonsense family; identity-by-descent context typical of recessive ultra-rare disease). Reported molecular lesions: homozygous nonsense mutation (PMID: 30528827) and homozygous 49.6-kb terminal-exon deletion (PMID: 37850634). **Environmental risk/modifiers:** mechanical trauma to mucosa (and skin) can precipitate or worsen erosions, consistent with mouse data showing traumatized skin develops suprabasilar acantholysis. **Protective factors:** none specifically established; conceptually, robust mucosal DSG1 expression would be protective (basis of skin sparing). **Gene–environment interaction:** trauma × DSG3-null state determines lesion extent (mucosa blisters spontaneously; skin requires added trauma).

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset / severity / frequency
Oral mucosal blisters/erosions	Clinical sign	HP:0000155 (Oral ulcer)	Congenital/neonatal–infantile; moderate–severe; characteristic (both families)
Laryngeal / respiratory mucosal friability & blistering	Clinical sign	HP:0001600 (Laryngeal abnormality)	Early onset; potentially severe (airway); prominent in 2024 twins
Suprabasal acantholysis on biopsy	Histopathologic	—	Diagnostic hallmark
Feeding difficulty / failure to thrive	Physical manifestation	HP:0011968 (Feeding difficulties); HP:0001508 (Failure to thrive)	Infancy; variable (strong in mouse model)
Hair anchoring defect (telogen)	Physical manifestation	HP:0001596 (Alopecia)	Documented in mouse; human extent uncertain

Quality-of-life impact: chronic oral pain, dysphagia/odynophagia, feeding difficulty, hoarseness/airway compromise, and risk of recurrent erosion-related infection; substantial impact on nutrition, speech, and airway safety. No formal EQ-5D/SF-36 data exist for this ultra-rare disease.

4. Genetic / Molecular Information

Causal gene: *DSG3* (desmoglein 3), 18q12.1, HGNC:3050, OMIM *605595*. **Pathogenic variants reported:** (1) homozygous **nonsense** variant (PMID: 30528827); (2) homozygous **structural deletion** — 49.6 kb at 18q12.1 removing the terminal *DSG3* exon (PMID: 37850634). **Classification:** pathogenic (ACMG: null variants in a gene where LOF is the mechanism). **Functional consequence:** complete loss of function (no DSG3 protein/adhesion). **Origin:** germline, biallelic (recessive). **Allele frequency:** both are private/ultra-rare; absent or vanishingly rare in gnomAD. **Modifier genes:** DSG1 (and other desmogleins/desmocollins)

plausibly modify tissue-specific severity via compensation, though not formally mapped in patients. **Epigenetic / chromosomal:** no epigenetic mechanism implicated; the 18q12.1 deletion is a small structural/CNV change detectable by CMA or sequencing-based CNV calling.

5. Environmental Information

ABOLM is fundamentally genetic. **Environmental factors** act only as modifiers: mechanical trauma to mucosa and skin precipitates erosions (mouse: traumatized skin shows suprabasilar acantholysis). No causative **toxin, radiation, lifestyle, or infectious agent** is implicated. N-glycosylation is required for desmoglein stability and desmosome formation (PMID: 30291216), indicating pharmacologic/metabolic perturbations of glycosylation could in principle aggravate desmosomal fragility, but this is not a disease trigger in ABOLM per se.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

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1. Biallelic LOF mutation in DSG3 (nonsense or terminal-exon deletion)
     └─ results in →
2. Absence of desmoglein-3 protein in stratified squamous epithelia
     └─ leads to →
3. Loss of the calcium-dependent N-terminal (EC1) trans-adhesive
   interface (V3/K7/P8/D59) between adjacent keratinocytes
     └─ results in →
4. Desmosomes in suprabasal/basal mucosal layers fail; desmosomes separate
   (EM: desmosome separation, "half-desmosomes")
     └─ BRANCH by tissue (desmoglein compensation theory) →
        4a. Oral/laryngeal mucosa: DSG1 weak → NO compensation →
            suprabasal acantholysis → mucosal blisters/erosions  [DOMINANT]
        4b. Skin: DSG1 broad → compensation → largely spared
            (acantholysis only with added trauma)                [SPARED]
        4c. Hair follicle outer root sheath (telogen): loss of
            anchoring → cyclical acantholytic hair shedding (mouse)
     └─ leads to →
5. Clinical manifestation: painful oral erosions, laryngeal/respiratory
   mucosal friability (airway risk), feeding difficulty/runting;
   histology identical to pemphigus vulgaris but WITHOUT autoantibodies
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Upstream vs downstream. Upstream (initiating): the *DSG3*-null genotype and absent adhesive interface. Downstream (effector): desmosome separation, acantholysis, epithelial fragility, and secondary consequences (pain, feeding failure, airway compromise). The **direction** of the ABOLM lesion is *loss of adhesion from the start* — contrasting with PV, where identical

downstream acantholysis is reached *from* autoantibody binding plus downstream signaling (RhoA inhibition and loss of desmosomal tension, PMID: 40740498; increased cortical myosin IIA and junctional remodeling upon desmosome disruption, PMID: 25006807). In ABOLM there is no antibody and no signaling trigger — the adhesion molecule is simply never made.

Molecular/cellular detail. Pathway: cell–cell adhesion via the desmosome (desmosomal cadherin–plakoglobin–desmoplakin–keratin axis). Cellular processes: loss of keratinocyte cell–cell adhesion (acantholysis), epithelial barrier failure. Suggested GO terms: **GO:0007156** (homophilic cell adhesion via plasma-membrane adhesion molecules), **GO:0002934** (desmosome organization), **GO:0030057** (desmosome, cellular component), **GO:0098609** (cell–cell adhesion). Protein dysfunction: complete loss of function of a desmosomal cadherin (not misfolding/aggregation — the protein is absent). Immune involvement: **none** (the defining distinction from autoimmune pemphigus). Metabolic changes: none primary; N-glycosylation is required for normal desmoglein stability/desmosome assembly (PMID: 30291216) but is not altered in ABOLM.

Cell types & anatomy: keratinocytes of oral and laryngeal stratified squamous epithelium (CL:0000312 keratinocyte); follicular outer-root-sheath keratinocytes (in mouse).

7. Anatomical Structures Affected

- **Organ level (primary):** oral cavity mucosa (UBERON:0000167 oral cavity; UBERON:0003729 mouth mucosa), larynx (UBERON:0001737), and respiratory mucosa (UBERON:0000483 epithelium). **Secondary:** nutritional/airway complications. **Body systems:** digestive (upper) and respiratory.
- **Tissue/cell level:** stratified squamous **epithelium** (UBERON:0002481); target cells = **keratinocytes** (CL:0000312) in basal/suprabasal layers; hair follicle outer root sheath (mouse).
- **Subcellular level:** the **desmosome** (GO:0030057) at the keratinocyte plasma membrane; adhesive failure localizes to the intercellular desmosomal space (EC1 interface); cytoplasmic plaque (plakoglobin/desmoplakin–keratin) secondarily disengages.
- **Localization / lateralization:** bilateral/diffuse mucosal involvement (not lateralized); mucosa-dominant with relative skin sparing.

8. Temporal Development

Onset: congenital/neonatal–infantile; mucosal erosions manifest early (the mouse becomes symptomatic within ~2–3 weeks of life; humans present in infancy/childhood with oral/airway disease). **Pattern:** chronic, lifelong. **Progression:** non-progressive at the genetic level but **episodic/fluctuating** clinically, with trauma-provoked flares and recurrent erosions; mouse hair loss is cyclical, tied to the telogen phase of the hair cycle. **Critical period:** infancy/early childhood (feeding and airway vulnerability) is the key window for supportive intervention. **Duration:** chronic, self-sustaining; no spontaneous remission of the underlying defect.

9. Inheritance and Population

Epidemiology: ultra-rare — only two families reported worldwide; prevalence/incidence not estimable (far below 1/1,000,000). **Inheritance:** autosomal recessive (*"only the second report in the literature of this unusual autosomal recessive blistering disorder,"* PMID: 37850634). **Penetrance:** appears complete for biallelic null genotype (both reported families affected). **Expressivity:** variable across the small sample (oral-dominant vs laryngeal/respiratory-dominant). **Consanguinity:** relevant (homozygous nonsense family). **Founder effects:** none established given only two private variants. **Carrier frequency:** not established; heterozygous carriers are expected to be asymptomatic (one functional *DSG3* allele suffices). **Demographics / sex ratio / geography:** too few cases to define; no sex predilection reported.

10. Diagnostics

Histopathology (biopsy): suprabasal/suprabasilar **acantholysis** with "tombstone" basal cells — indistinguishable from pemphigus vulgaris on H&E. **Immunofluorescence/serology — the decisive discriminator:** DIF and IIF **negative**; DSG1/DSG3 autoantibody **ELISA negative** (no autoantibodies). **Electron microscopy:** desmosome separation / "half-desmosomes." **Genetic testing (confirmatory):** biallelic *DSG3* loss-of-function. Recommended approach: single-gene *DSG3* sequencing or a blistering/epidermolysis gene panel; **chromosomal microarray (CMA) or CNV-aware sequencing** is required to detect structural deletions such as the 49.6-kb 18q12.1 deletion (PMID: 37850634). WES/WGS are useful when panels are negative or to detect CNVs. **Differential diagnosis:** pemphigus vulgaris (autoantibody-positive), paraneoplastic pemphigus, epidermolysis bullosa, Hailey-Hailey/Darier disease, and other acantholytic dermatoses — all distinguished by immunofluorescence status, autoantibody profile, and *DSG3* genetics. **Screening:** cascade/carrier testing within affected families; prenatal/preimplantation testing possible once the familial variant is known.

11. Outcome / Prognosis

Survival/mortality: no formal survival data; the chief life-threatening risk is **airway compromise** from laryngeal/respiratory mucosal friability, plus nutritional failure from oral disease (mouse runting is a direct analog). **Morbidity:** chronic oral pain, dysphagia/odynophagia, feeding difficulty, hoarseness, and recurrent mucosal erosions dominate. **Disease course:** chronic/lifelong with trauma-provoked flares. **Recovery potential:** the underlying genetic defect is not reversible with current therapy; supportive care can substantially mitigate morbidity. **Prognostic factors:** extent of laryngeal/respiratory involvement (airway risk) and feeding status are the key determinants of outcome. **Prognostic biomarkers:** none established beyond genotype.

12. Treatment

Management is supportive — NOT immunosuppressive. Because ABOLM is non-autoimmune, the pemphigus therapeutic armamentarium (systemic corticosteroids, rituximab, other immunosuppressants) is **not indicated**: *"The exclusion of an autoimmune basis has management implications"* (PMID: 37850634). Core supportive measures:

Domain	Intervention	NCIT suggestion
Airway	Airway monitoring/protection; ENT/otolaryngology surveillance; intervention for laryngeal friability	NCIT:C15329 (Supportive Care)
Nutrition	Soft/modified diet, feeding support, nutritional supplementation	NCIT:C15311 (Nutritional Support)
Pain / mucosa	Topical analgesia, oral care, antiseptics, management of secondary infection	NCIT:C15329 (Supportive Care)
Trauma avoidance	Minimizing mucosal mechanical trauma; dental/oral care precautions	—

Advanced/experimental (none approved; conceptual): gene-/protein-replacement strategies aimed at restoring DSG3 are biologically rational but untested; no clinical trials exist for this ultra-rare disease. **Pharmacogenomics:** not applicable. The key treatment principle for the knowledge base is the **active de-adoption** of immunosuppression in favor of supportive airway/nutrition/trauma care.

13. Prevention

Primary prevention is genetic: **genetic counseling** for consanguineous/at-risk families, carrier testing, and reproductive options (prenatal diagnosis, preimplantation genetic testing) once the familial *DSG3* variant is identified. **Secondary prevention:** early diagnosis to redirect management away from unnecessary immunosuppression and toward airway/nutritional surveillance. **Tertiary prevention:** minimizing mucosal trauma, aggressive oral hygiene/infection control, and proactive airway management to prevent complications. **Immunization/public-health/environmental interventions:** not applicable (non-infectious, non-environmental disease).

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** human *DSG3* (NCBI Gene 1830); mouse *Dsg3* (NCBI Gene 13512). Desmoglein 3 is evolutionarily conserved across mammals.
- **Natural disease:** the spontaneous mouse mutant **"balding" (bal)** is a naturally occurring *Dsg3*-null allele with a phenotype *"identical"* to the engineered knockout (PMID: 9166409) — the closest natural animal counterpart of ABOLM.
- **Comparative biology:** the mouse reproduces the human oral acantholytic histology (suprabasilar acantholysis, basal tombstoning, desmosome separation), confirming deep conservation of DSG3 adhesive function; mice additionally show prominent cyclical telogen hair loss, a feature less characterized in humans.
- **Zoonotic potential:** none (non-transmissible genetic disorder).

15. Model Organisms

Model	Type	Recapitulation	Reference
Dsg3−/− knockout mouse	Mammalian, genetic KO	Oropharyngeal suprabasilar acantholysis + tombstoning; desmosome separation (EM); runting from feeding failure; traumatized-skin acantholysis; cyclical telogen hair loss	PMID: 9166409; PMID: 9701552
"Balding" (*bal*) mouse	Spontaneous *Dsg3*-null allele	Identical to engineered KO	PMID: 9166409
AK23 anti-Dsg3 mAb (phenocopy)	Antibody-induced in vitro/in vivo	Reproduces PV/Dsg3-null phenotype by blocking the EC1 adhesive interface	PMID: 12574390

Applications: the *Dsg3*-null mouse enables study of desmoglein adhesion biology, acantholysis mechanisms, the desmoglein compensation theory, and hair-cycle–dependent adhesion. **Limitations:** the mouse's prominent hair phenotype may over-represent a feature that is minor in humans; human airway-dominant disease (laryngeal friability) is not a primary focus of the published mouse studies. Databases: MGI (mouse *Dsg3*), IMPC/IMSR for availability.

Mechanistic Model / Interpretation

ABOLM should be understood as **"pemphigus vulgaris minus the autoantibody."** Pemphigus and ABOLM converge on an identical effector lesion — suprabasal acantholysis from DSG3 adhesive failure — but arrive there from opposite directions:

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PEMPHIGUS VULGARIS (acquired, autoimmune)
  Loss of immune tolerance → anti-DSG3 (±anti-DSG1) IgG
     → antibody binds EC1 interface / steric + signaling (RhoA↓, tension loss)
        → desmosome disassembly → suprabasal acantholysis

ABOLM (congenital, genetic)
  Biallelic DSG3-null genotype → no DSG3 protein at all
     → EC1 adhesive interface never forms
        → desmosome adhesion absent → suprabasal acantholysis
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Both are then **filtered by the same tissue rule** — the desmoglein compensation theory. Wherever DSG1 can substitute (most skin), the lesion is masked; wherever it cannot (oral and laryngeal mucosa; telogen follicle), the lesion is expressed. This is why an autoimmune disease and a recessive genetic disease share the same target-tissue signature. The genetic disease is, in a sense, the cleanest possible proof of the compensation model: it removes the confound of variable autoantibody titers and shows that *absence of DSG3* alone, interpreted through tissue-specific DSG1 availability, predicts exactly where blisters form.

The practical corollary — and the single most important clinical takeaway — is **therapeutic**: the two diseases look identical on H&E but require opposite management. Pemphigus demands immunosuppression; ABOLM would be *harmed* by it (immunosuppression cannot replace a missing adhesion protein and exposes an infant to needless risk). The diagnostic fork is immunofluorescence/serology (negative in ABOLM) plus *DSG3* genetics.

Evidence Base

PMID	Title (abbrev.)	Evidence type	How it supports the findings
37850634	Homozygous *DSG3* terminal-exon deletion in ABOLM (twins)	Human genetic	Second family; defines ABOLM; AR inheritance; non-autoimmune; laryngeal/respiratory phenotype; management implications
30528827	Homozygous nonsense *DSG3* mutation causing oral/laryngeal acantholytic blisters	Human genetic	First family; establishes LOF *DSG3* as cause
9166409	*Dsg3* knockout mouse (PV-like)	Model organism	Recapitulates oral acantholytic histology; proves DSG3 adhesion; "balding" = natural null
9701552	DSG3 anchors telogen hair	Model organism	Telogen-specific follicular adhesion role; acantholysis in outer root sheath
12574390	AK23 induces pemphigus by targeting EC1 interface	In vitro / structural	Localizes adhesion to calcium-dependent N-terminal interface (V3/K7/P8/D59)
15858475	Cutaneous-type PV	Human clinical	States the desmoglein compensation theory
12677426	Pathophysiology of pemphigus (review)	Review	Compensation theory explains oral-vs-skin localization
12366435	Neonatal PV from mucosal-dominant mother	Human clinical	Clinical evidence for Dsg compensation theory
40740498	RhoA/tension loss in PV	In vitro	Contrast: PV downstream signaling (absent in genetic ABOLM)
25006807	Adhesion/contractility upon desmosome disruption	In vitro	Downstream consequences of losing desmosomal adhesion
30291216	N-glycosylation and desmosome stability	In vitro	Desmoglein biology; potential aggravating factor (not a trigger)
23368972	Desmosomal adhesion & PV (review)	Review	Context: DSG3 cloning, KO, mAbs clarified PV pathogenesis

PMID	Title (abbrev.)	Evidence type	How it supports the findings
38067138	DSG3 as potential oncogene in oral cancer	In vitro	Broader DSG3 biology in oral epithelium (tangential)

Consistency of evidence: 2 of 2 human families show biallelic *DSG3* LOF with the same core phenotype; 1 engineered + 1 spontaneous mouse null reproduce the oral acantholytic histology; the structural/antibody data independently localize the critical adhesive function to the DSG3 EC1 interface. The causal chain is thus supported convergently across human genetic, model-organism, and in-vitro/structural evidence.

Limitations and Knowledge Gaps

1. **Extreme rarity (n ≈ 2 families).** All human conclusions rest on two reports; penetrance, expressivity range, natural history, prognosis, and sex/geographic patterns cannot be quantified.
2. **Incomplete abstract for the index family.** PMID: 30528827 has no available abstract; details (exact variant nomenclature, full phenotype) are drawn from the disease definition and corroborating report rather than a quoted abstract.
3. **Human hair/skin phenotype under-characterized.** The telogen hair-anchoring defect and trauma-induced skin acantholysis are well documented in mouse; their human counterparts are not clearly described, leaving uncertainty about the full human phenotypic spectrum.
4. **No natural-history, QoL, survival, or treatment-outcome data.** Airway risk is inferred from the laryngeal/respiratory phenotype; no formal mortality or morbidity statistics exist.
5. **No modifier-gene mapping in patients.** DSG1's compensatory role is inferred from the compensation theory and PV data, not directly measured in ABOLM patients.
6. **No approved or trialed disease-specific therapy.** Management is supportive by default; no gene/protein-replacement data exist.

Proposed Follow-up Experiments / Actions

1. **Establish an international ABOLM/*DSG3*-null registry** to aggregate cases, variants, natural history, airway outcomes, and QoL — the only path to epidemiologic and prognostic estimates for an n≈2 disease.
2. **Characterize the human skin and hair phenotype systematically** (trauma-provoked acantholysis; telogen hair fragility) to test whether mouse features translate to humans.
3. **Quantify DSG1 expression in patient oral/laryngeal vs skin biopsies** (IHC/transcriptomics) to directly validate the desmoglein compensation theory as the explanation for ABOLM localization.
4. **Diagnostic guidance:** codify a decision pathway — PV-like acantholysis + negative DIF/IIF/ELISA → *DSG3* sequencing **plus CMA/CNV analysis** (to avoid missing structural deletions like the 18q12.1 event).
5. **Airway surveillance protocol:** given laryngeal/respiratory friability, define ENT monitoring and intervention thresholds to reduce airway morbidity.
6. **Proof-of-concept DSG3 restoration** in *Dsg3*-null keratinocytes/organoids and the *bal* mouse (gene/mRNA/protein replacement) as a preclinical foundation for any future therapy.
7. **Genetic counseling pathway** for consanguineous and affected families, including carrier testing and reproductive options once the familial variant is known.

Report compiled from 6 confirmed findings and 18 reviewed papers over 5 investigation iterations. Evidence spans human clinical/genetic, model-organism (mouse), and in-vitro/structural sources, as indicated per finding.
